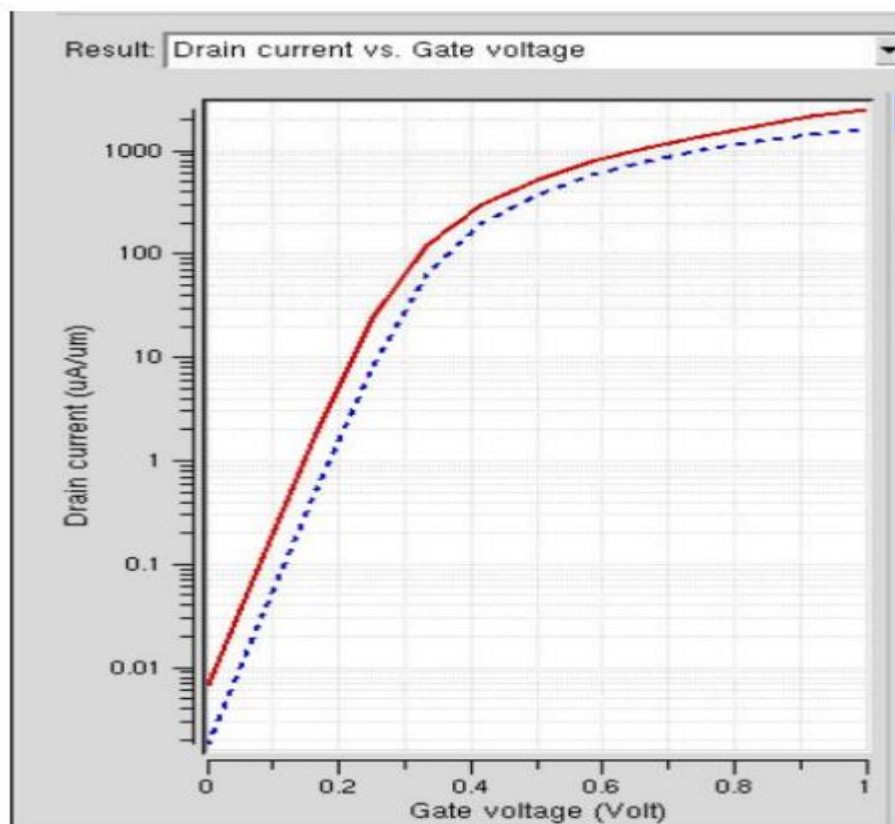


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Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

CASE 1: increase in V_g :

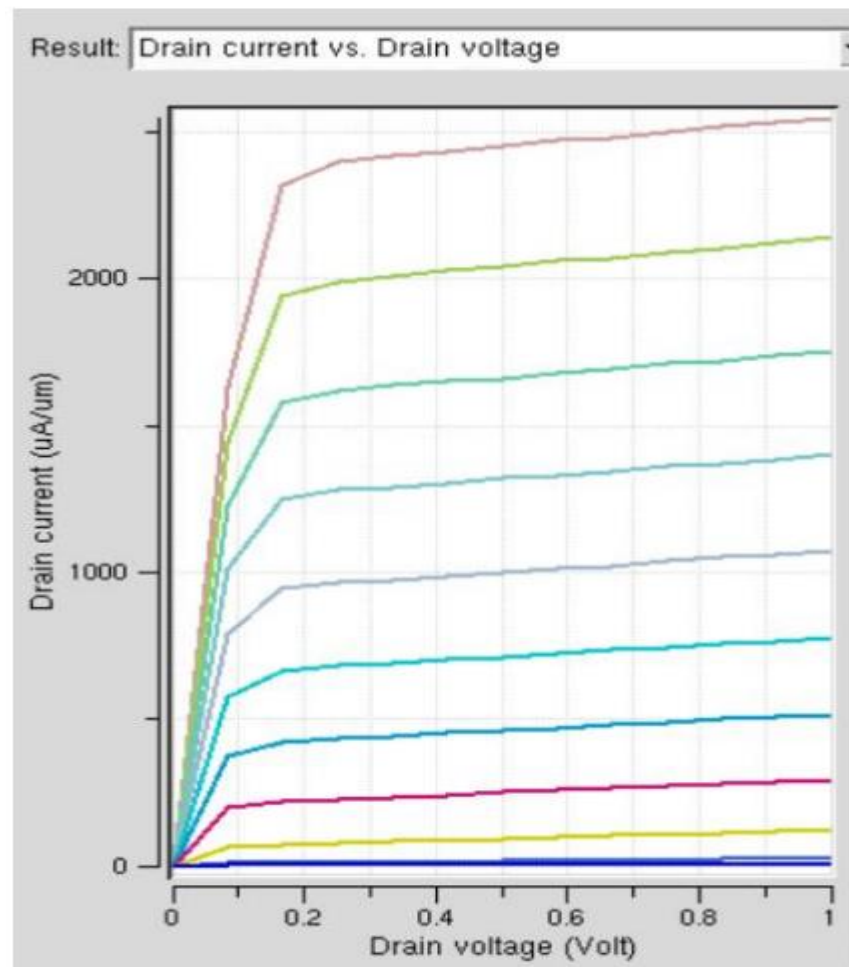
observation :As V_g increase,the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.

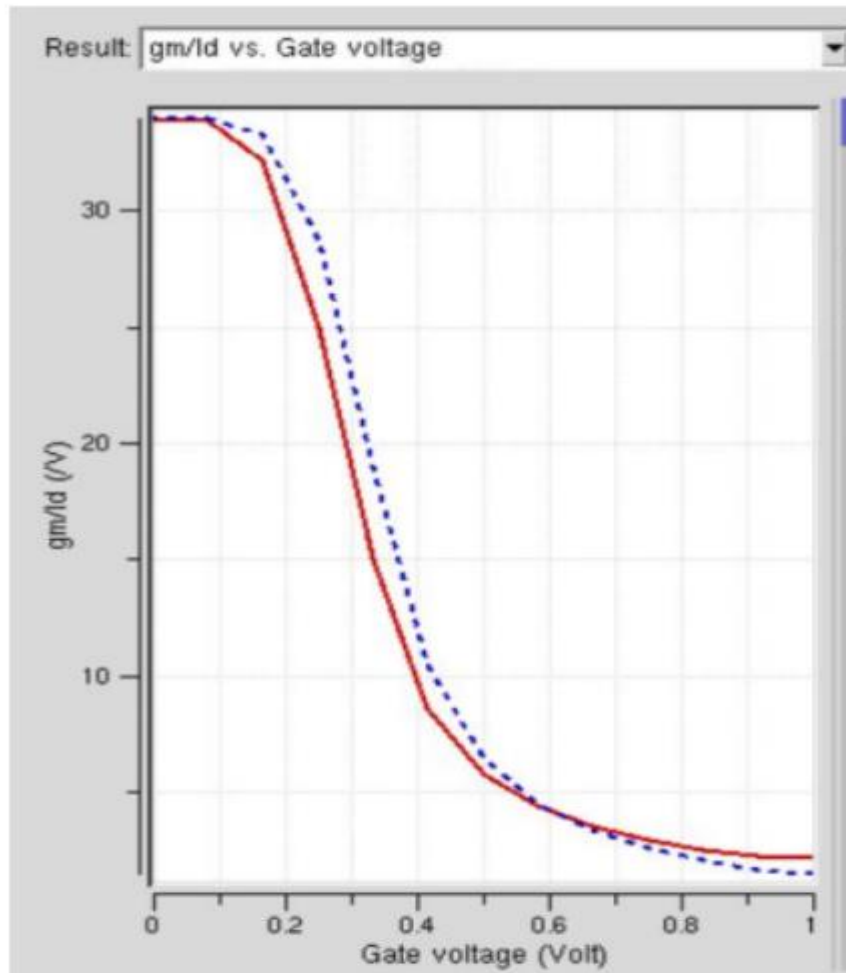


The objective was to study the effect of increasing the gate voltage (V_g) on the drain current (I_D) using the NanoHUB online simulation tool. The simulation was performed by gradually increasing V_g while keeping the other parameters constant. The purpose of this study was to observe how gate voltage controls the conduction of the MOSFET. As V_g increases, the drain current (I_D) increases rapidly because a stronger inversion channel is formed, causing the MOSFET to enter the saturation region earlier. The simulation demonstrates the relationship between gate voltage and drain current, highlighting the switching behavior of the MOSFET.

CASE 2: variation in V_D :

Observation : drain current (I_D) saturates for higher v_d at a fixed V_g





The objective was to study the effect of **varying the drain voltage (V_D)** on the **drain current (I_D)** using the **NanoHUB online simulation tool**. The simulation was performed by increasing V_D while keeping the **gate voltage (V_G)** constant. The purpose of this study was to observe the output characteristics of the MOSFET. As V_D increases, the **drain current (I_D)** initially increases and then **saturates at higher drain voltages** due to the MOSFET entering the **saturation region**. The simulation demonstrates the relationship between **drain voltage** and **drain current**, highlighting the saturation behavior of the MOSFET.

Conclusion:

The simulation shows that the **drain current (I_D)** increases with **gate voltage (V_G)** and **saturates at higher drain voltages (V_D)**. This confirms the **input and output characteristics of the MOSFET** and demonstrates its operation in the **saturation region**.